Appendix 2: Lake Washington Analysis

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Here we show the GAM and DLM analysis for the Lake Washington dataset. For complete code, see the full quarto document. We include some code snippets here for reference.

Data Processing

In Lake WA the major change ecosystem change is related to sewage in the lake – adding wastewater plants in the 1960s reduced phosphorous (Figure S1) and in turn, blue green algae (Figure S2) which can be seen by examining time series plots of the data.

We can then use the MARSS package to fit a DLM. We'll filter the data to only use counts before 1974, and log transform densities. We also z-score the covariate. This is a good example of imperfect data – there's a couple of missing observations in the covariate, which we can't have in a DLM or GAM – so let's just fill them in with a spline. We can do this as follows:

```
dat <- dplyr::filter(dat, Year < 1974) %>%
  dplyr::mutate(y = log(Bluegreens), zp = as.numeric(scale(TP)))

interpolated <- spline(dat$zp, xout = 1:nrow(dat))$y
dat$zp[which(is.na(dat$zp))] <- interpolated[which(is.na(dat$zp))]
```

The response data exhibit a strong seasonal cycle – and it's a good idea to de-season that to focus on the phosphorous effect. Here we de-season the data:

```
# Fill in a few missing values
dat$y_interp <- dat$y
interpolated <- spline(dat$y, xout = 1:nrow(dat))$y
missing <- which(is.na(dat$y_interp))
dat$y_interp[missing] <- interpolated[missing]
```

```
dat <- dplyr::group_by(dat, Month) %>%
  dplyr::mutate(month_mean = mean(y,na.rm=T)) %>%
  dplyr::ungroup() %>%
  dplyr::mutate(y_adj = y_interp - month_mean)
```

Last, we should look at the lags between driver and response relationships which we can do using a CCF (Figure S3). The CCF here indicates a 2-4 month lag, where TP and algae are more strongly associated with one another.

We'll use a lag of 4 months, which we can apply as follows:

```
lag_n <- 5
dat$lagged_zp <- c(rep(NA, lag_n), dat$zp[1:(nrow(dat)-lag_n)])
dat <- dat[-c(1:lag_n),]
```

DLMs

First, we need to define the model – this block is basically copied from the MARSS book, the salmon survival case study

```
m <- 2
TT <- nrow(dat)
B <- diag(m) ## 2x2; Identity
U <- matrix(0, nrow = m, ncol = 1) ## 2x1; both elements = 0
Q <- matrix(list(0), m, m) ## 2x2; all 0 for now
diag(Q)[1] <- 0.0000001
diag(Q)[2] <- c("q.beta")
#diag(Q) <- c("q.alpha", "q.beta") ## 2x2; diag = (q1,q2)
Z <- array(NA, c(1, m, TT)) ## NxMxT; empty for now
Z[1, 1, ] <- rep(1, TT) ## Nx1; 1's for intercept
Z[1, 2, ] <- dat$lagged_zp ## Nx1; predictor variable
A <- matrix(0) ## 1x1; scalar = 0
R <- matrix("r") ## 1x1; scalar = r
## only need starting values for regr parameters
inits_list <- list(x0 = matrix(c(0, 0), nrow = m))
## list of model matrices & vectors
mod_list <- list(B = B, U = U, Q = Q, Z = Z, A = A, R = R)
# convert response to matrix
dat_mat <- matrix(dat$y_adj, nrow = 1)
# fit the model -- crank up the maxit to ensure convergence
```

```r
dlm_1 <- MARSS(dat_mat, inits = inits_list, model = mod_list,
control = list(maxit=4000), method="TMB")
```

```
MARSS fit is
Estimation method: TMB
Estimation converged in 18 iterations.
Log-likelihood: -155.082
AIC: 318.1641   AICc: 318.4626

          Estimate
R.r          0.226
Q.q.beta     0.235
x0.X1        0.498
x0.X2        1.422
Initial states (x0) defined at t=0

Standard errors have not been calculated.
Use MARSSparamCIs to compute CIs and bias estimates.
```

Let's put the state estimates back into the original dataframe and do some plotting. The slope is generally positive, which is what we expect for the hypothesized relationship between bluegreen algae and total phosphorus – and exhibits clear variation through time (Figure S4). We can also plot the fitted values from our model (Figure S5).

And we can examine the residuals for a temporal trend, and see that because the DLM is flexible, there's not much evidence of a trend (Figure S6) and there are no issues wiht the q-q plot (Figure S7).

GAMs

We can fit a GAM model 5 ways, 1) so that the smoother is a function of phosphorus which represents a nonlinear functional relationship between total phosphorus and bluegreen algae that is not temporally structured thus the relationship is stationary and fixed through time. This is how GAMs are typically structured for ecological analyses. 2) a temporally structured model such that the abundance of bluegreen algae varies as a function of time unrelated to phsophorus, 3) A temporally structured model so that the smooth function of phosphorus varies by time, and 4) a smooth on date as a driver of cyanobacteria where the smooth through time varies by phosphorus level. Finally, these relationships can be fit as 5) a tensor product which provide a multivariate function spanning the dimensions of time and phosphorus. We will not discuss tensor products in greater detail and only include them for completeness.

We show the model predicted cyanobacteria trend through time using these five modeling approaches (Figure S8). While they are quite similar, the predictions diverge sustantially at the end of the time series.

```
dat$n_date <- as.numeric(as.factor(dat$date))

gam_cr1 <- gam(y_adj ~ s(lagged_zp, bs = "cr"), data = dat) #1
gam_cr2 <- gam(y_adj ~ s(lagged_zp, by = n_date, bs = "cr"), data = dat) #2
gam_cr3 <- gam(y_adj ~ s(n_date, by = lagged_zp, bs = "cr"), data = dat) #3
gam_cr4 <- gam(y_adj ~ s(n_date), data = dat) #4
gam_cr5 <- gam(y_adj ~ te(lagged_zp, n_date), data = dat) #5
```

```
[1] "#26172AFF" "#414081FF" "#357BA2FF" "#3DB4ADFF" "#A8E1BCFF"
```

It is important to also examine the residuals from these models. For example, we can look at the residuals for the GAMs fit to the Lake Washington data. In looking across models, those that do not include time-varying parameters show evidence of a persistent trend in the residuals that is not accounted for by the model ((Figure S9) despite teh q-q plot being reasonable for all model specifications ((Figure S10).

```
resid_df_1 <- data.frame(date = dat$date, resid = resid(gam_cr1),
                         Model = "s(Phos)")
resid_df_2 <- data.frame(date = dat$date, resid = resid(gam_cr2),
                         Model = "s(Phos, by = date)")
resid_df_3 <- data.frame(date = dat$date, resid = resid(gam_cr3),
                         Model = "s(date, by=Phos)")
resid_df_4 <- data.frame(date = dat$date, resid = resid(gam_cr4),
                         Model = "s(date)")
resid_df_5 <- data.frame(date = dat$date, resid = resid(gam_cr5),
                         Model = "te(Phos, date")
resid <- rbind(resid_df_1, resid_df_2, resid_df_3, resid_df_4, resid_df_5)
gam_resid<-ggplot(resid, aes(date, resid, group = Model)) +
  geom_point() +
  geom_smooth() + theme_classic() +
  facet_wrap(~ Model) + xlab("Date") +
  ylab("Residuals") + ggtitle("GAM Residuals")
```

There are a few more analysis approaches for running GAMs including the choice of smoother function (Figure S8 & Figure S11) and the number of knots, or the "wiggliness" (Figure S12) of the smoothed relationship. We can plot these different results and include the results of the DLM for comparison, note that 'cr' and 'tp' are so similar they are difficult to discern from the plot. We can do a similar comparison for the choice of k. Here we compare a four different numbers of knots (Figure S12), we show an intentionally wide range of k (30 - 120) to illustrate the effects of k on estimated uncertainty, in practice k should be specified based on degrees of freedom and expectations of variability through time.

Figures

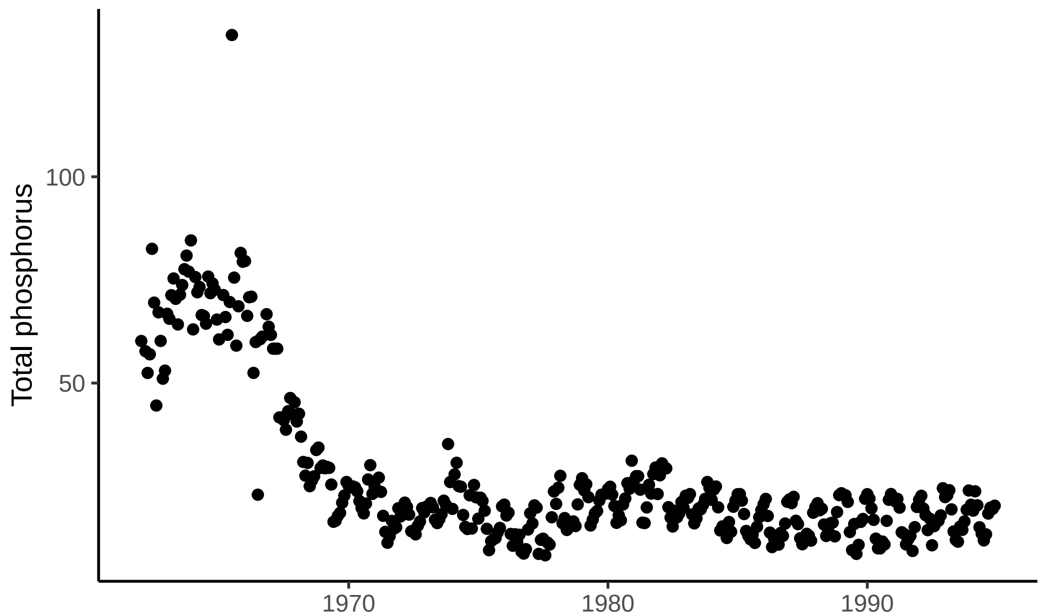

Figure S1: Time series of total phosphorus in Lake Washington

Figure S2: Time series of bluegreen algae (cyanobacteria) abundance in Lake Washington

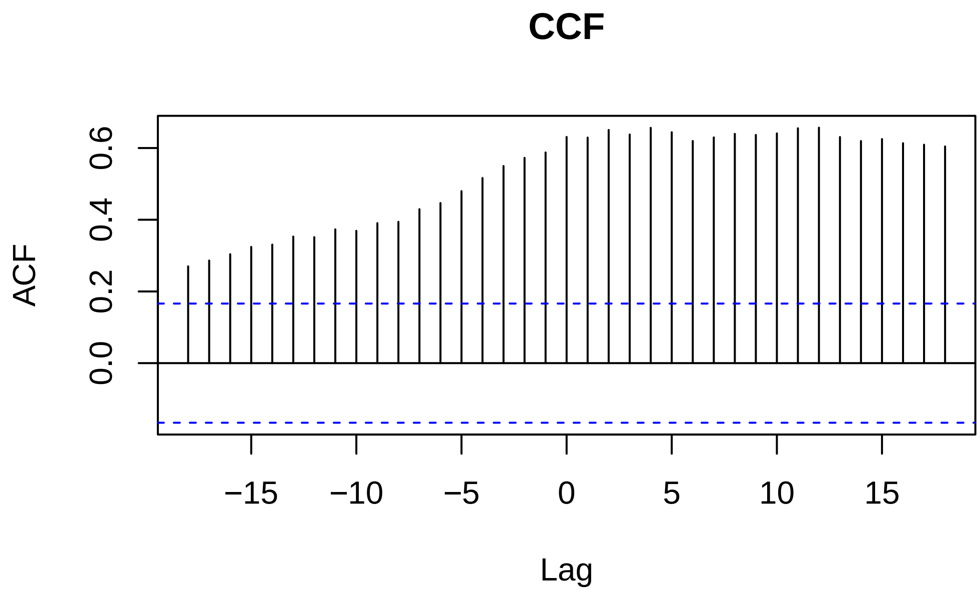

Figure S3: A CCF (Cross-Correlation Function) plot that represents the correlation between total phosphorus and blugreen algae abundance as a function of the lag (time difference) between them.

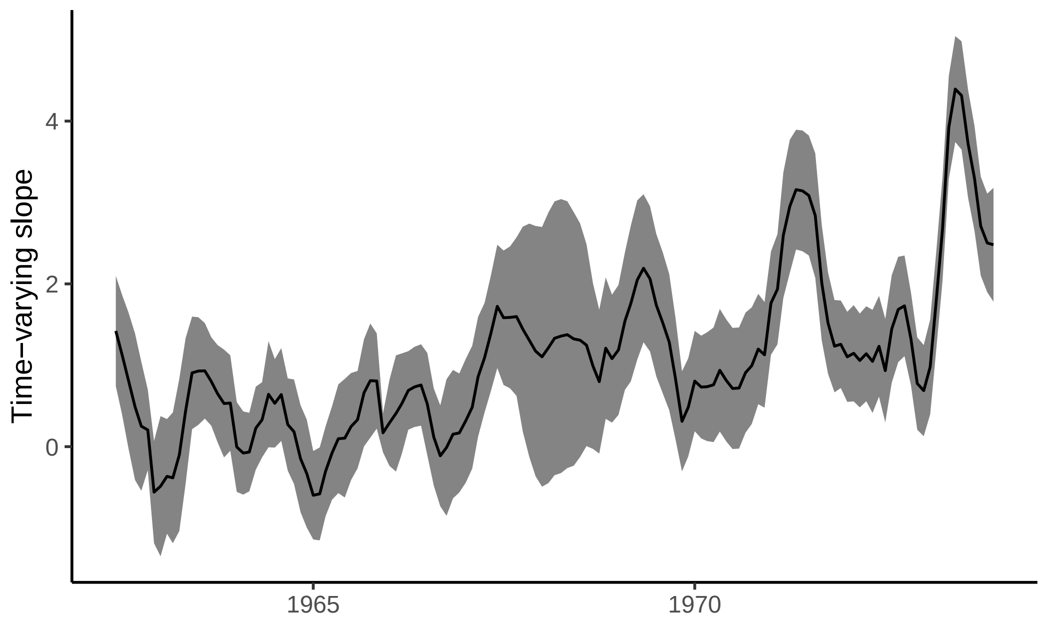

Figure S4: Time-varying slope of the relationship between bluegreen algae (response) and total phosphorus (driver) from a DLM

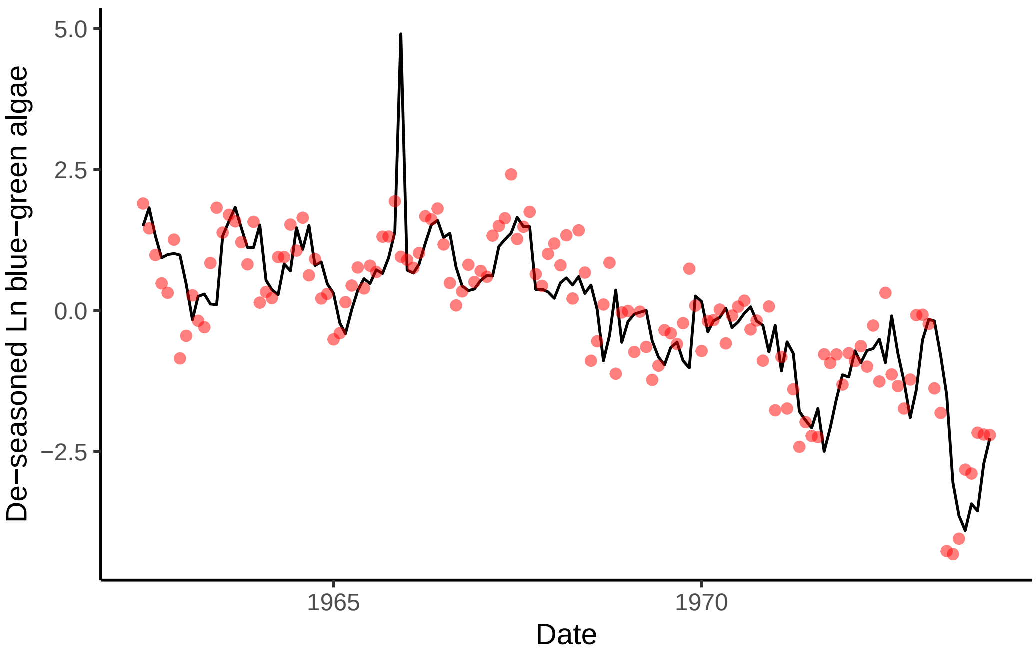

Figure S5: Fit of the DLM model with time-varying slope (black line) to observed bluegreen algae abundance (red points).

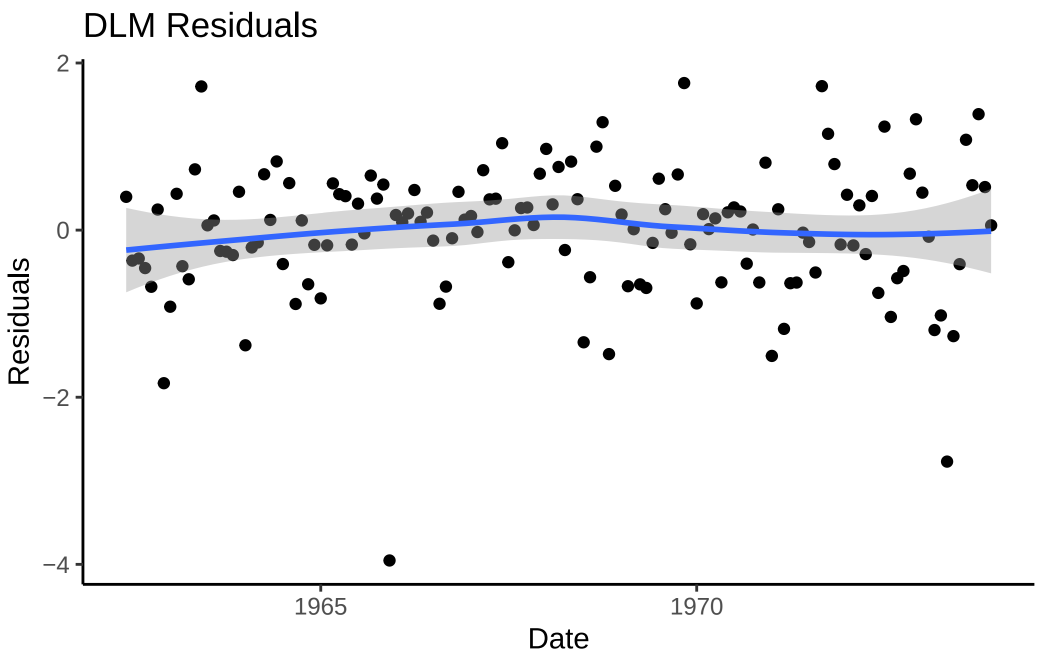

Figure S6: Time series of residuals from the DLM model fit with a time-varying slope between blue-green algae and phosphorus

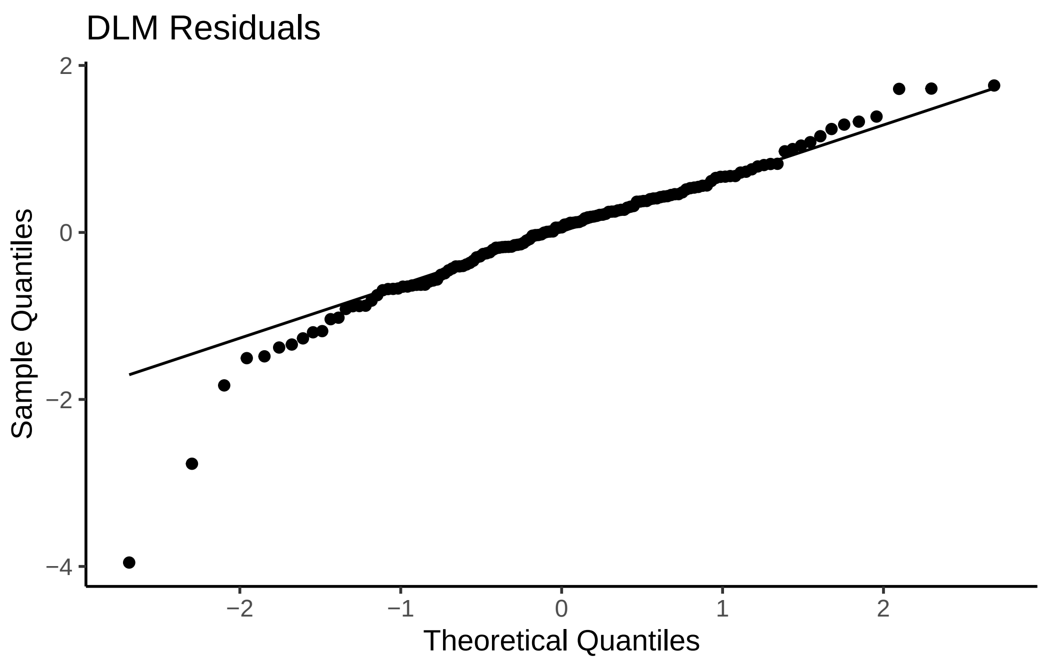

Figure S7: Q-Q plot from the DLM model fit with a time-varying slope between blue-green algae and phosphorus

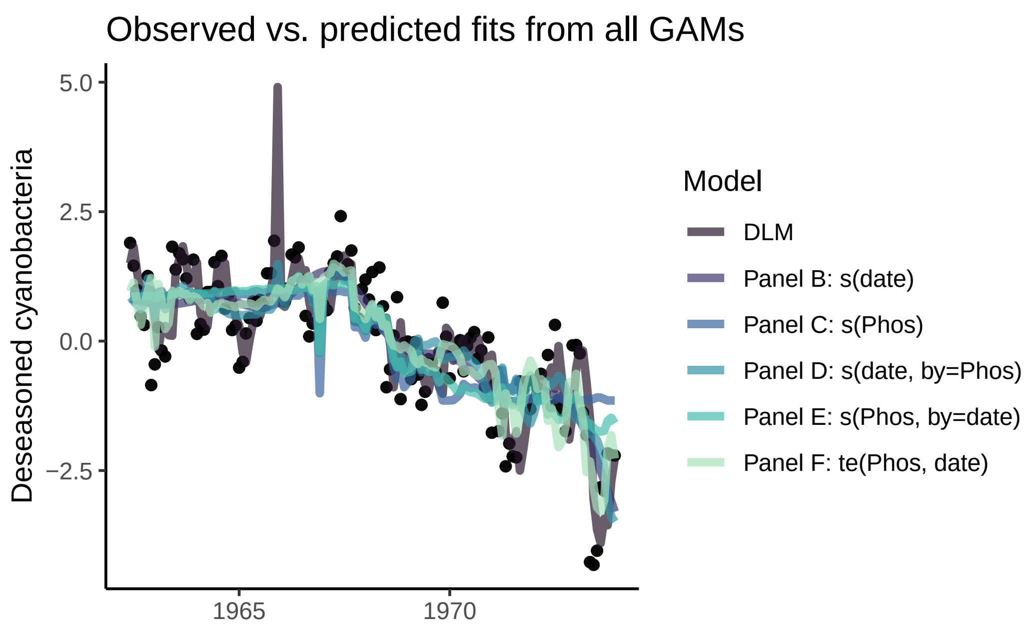

Figure S8: Fit of the GAM models (lines) to observed bluegreen algae (cyanobacteria) abundance (black points). Colors denote different specifications of the smoother functions.

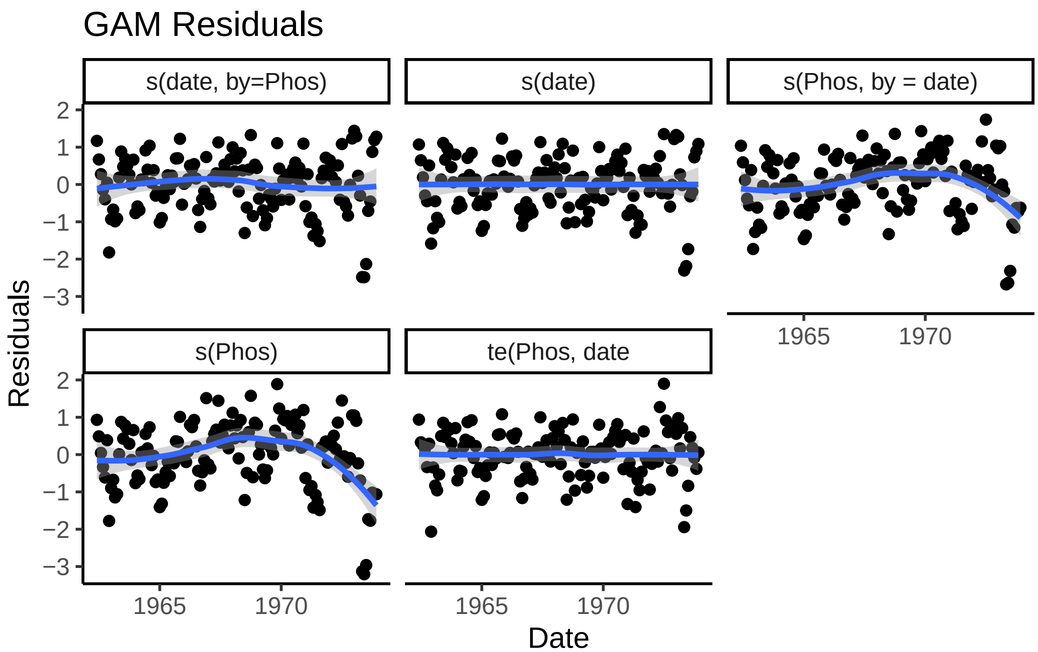

Figure S9: Time series of residuals from the five GAM models where residuals from models that do not have a temporally structured smooth term (s(Phos) and s(Phos, by = date)) have a trend through time.

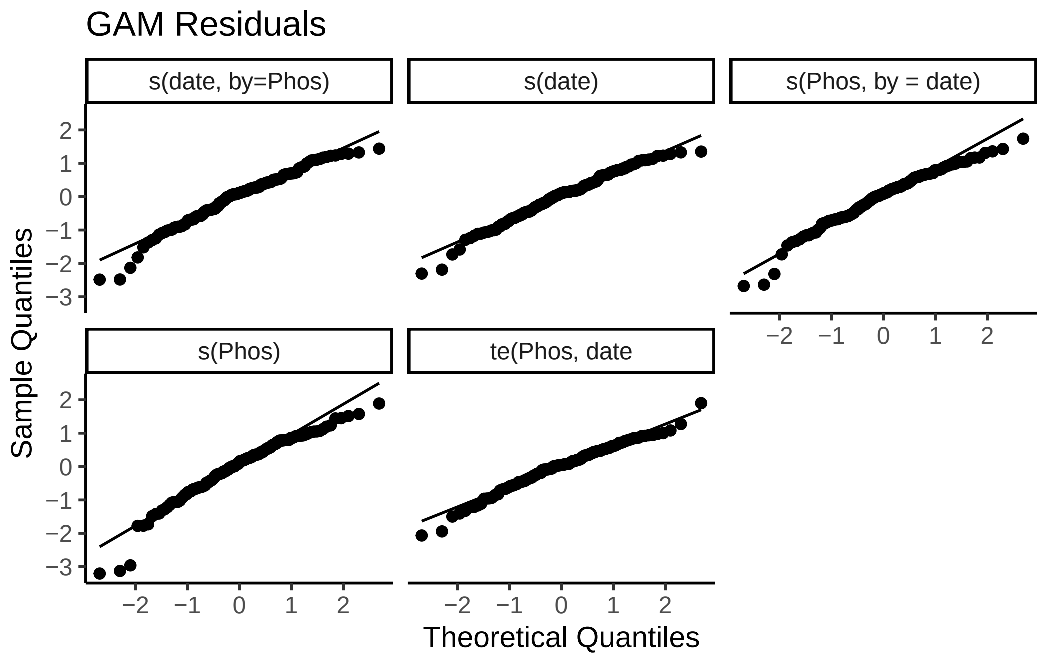

Figure S10: Q-Q for each of the five GAM models where residuals from models that do not have a temporally structured smooth term (s(Phos) and s(Phos, by = date)) have a trend through time.

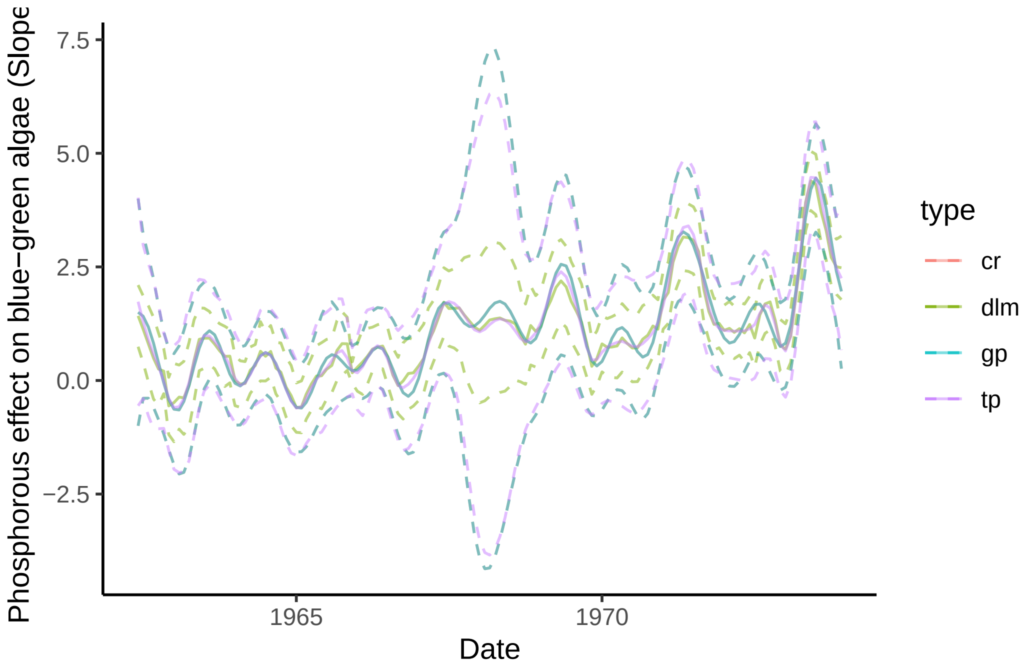

Figure S11: Effect of the choice of smoother function on estimates of time-varying slope parameter where color denotes model type. 3 GAM smoother types (cr, tp, gp) are contrasted with a DLM (dlm).

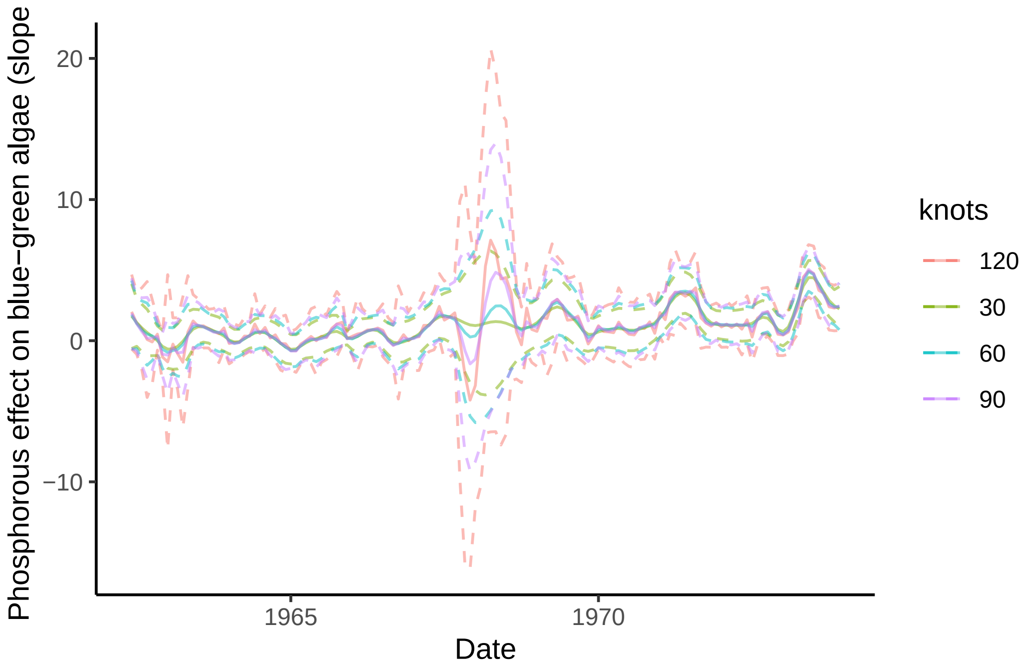

Figure S12: Effect of smoothing degrees of freedom on the standard errors of the estimates of time-varying slope parameter where color denotes model type. We show 4 distinct knots (colors)